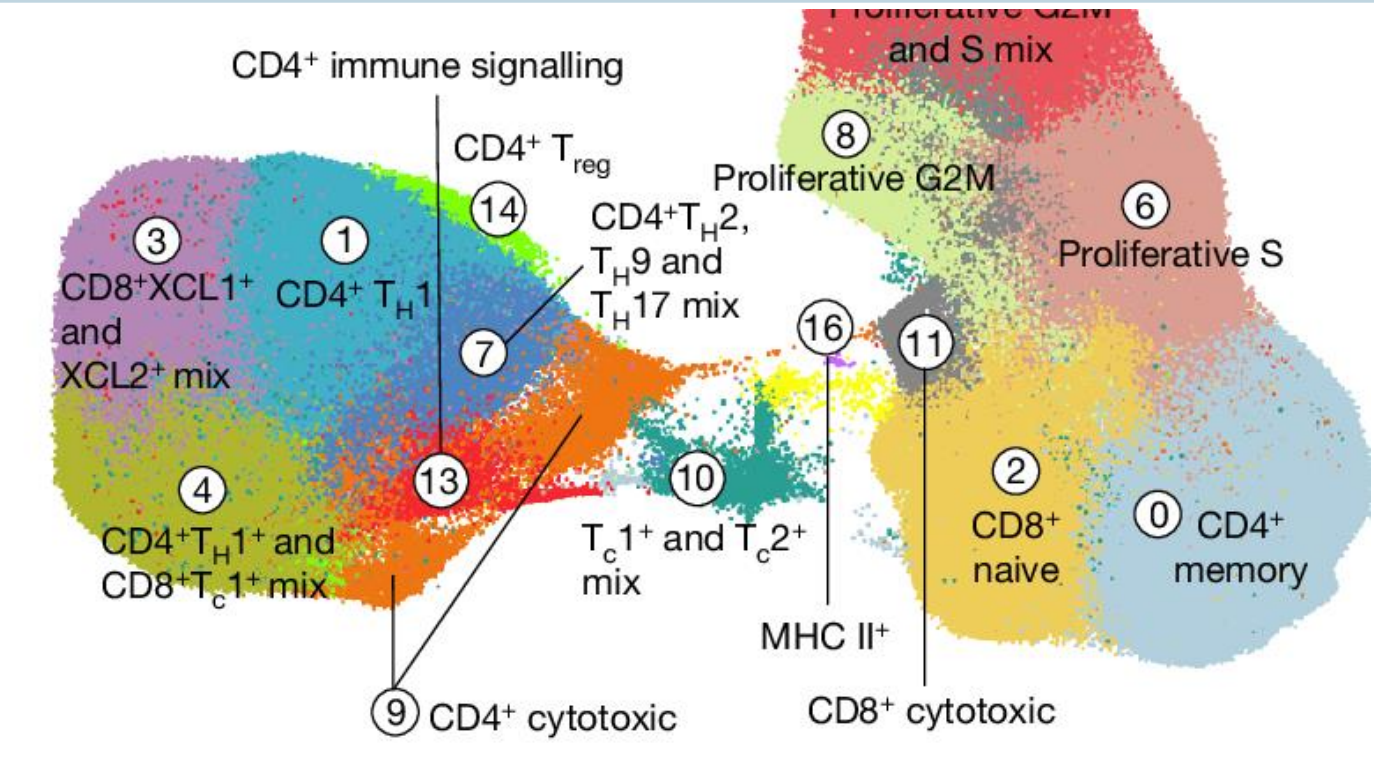


The atlas resolves one cohort into 17 measurable cell states

The UMAP is not the conclusion; it is the map used by every later comparison.



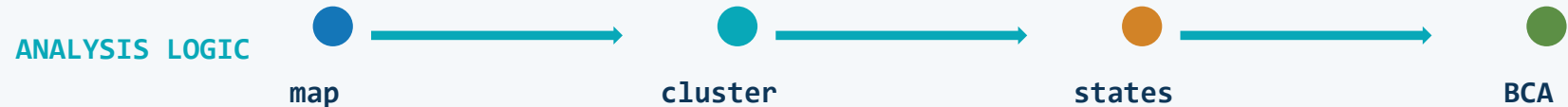
Source: Bai et al., Fig. 1b • selected UMAP / cluster annotation

HOW TO READ THE MAP

Each numbered region is a cell state.
Spatial proximity suggests a state landscape—not a causal pathway.

FOUR STATE FAMILIES

- NAIVE** memory / early
- CYTOTOXIC** effector program
- REGULATORY** signalling program
- PROLIFERATIVE** expansion program



17
cell states